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AI technology to support adaptive functioning in neurodevelopmental conditions in everyday environments: a systematic review

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Supports for adaptive functioning in individuals with neurodevelopmental conditions (NDCs) is of utmost importance to long-term outcomes. Artificial intelligence (AI)-assistive technologies has enormous potential to offer efficient, cost-effective, and personalized solutions to address these challenges, particularly in everyday environments. This systematic review examines the existing evidence for using AI-assistive technologies to support adaptive functioning in people with NDCs in everyday settings. Searches across six databases yielded 15 studies meeting inclusion criteria, focusing on robotics, phones/computers and virtual reality. Studies most frequently recruited children diagnosed with autism and targeted social skills (47%), daily living skills (26%), and communication (16%). Despite promising results, studies addressing broader transdiagnostic needs across different NDC populations are needed. There is also an urgent need to improve the quality of evidence-based research practices. This review concludes that AI holds enormous potential to support adaptive functioning for people with NDCs and for personalized health support. This review underscores the need for further research studies to advance AI technologies in this field.

Artificial intelligence (AI)-assistive devices have the potential to revolutionize healthcare for people with neurodevelopmental conditions (NDCs). Global estimates suggest that over 1 billion people live with some form of disability¹. This includes developmental disabilities such as autism spectrum disorder (ASD), intellectual disabilities (ID) and other conditions that emerge in early childhood¹. Prevalence rates vary across studies but have generally shown an increasing trend in numbers of affected individuals in recent years, explained in part to increased awareness of these conditions and changing diagnostic frameworks². For example, according to the Centers for Disease Control and Prevention (CDC), autism diagnoses have increased by 316% in the United States in the past 20 years². This growth has been recognized by leading international organizations including the World Health Organization (WHO) and the United Nations (UN), who have called for more actionable supports that facilitate opportunities for people with different disabilities to thrive in daily life^{1,3}. This increase in the prevalence of NDCs adds pressure to already strained systems. It places escalating demands on diagnostic and assessment services, adding to prolonged waiting times that hinder timely diagnosis and access to critical supports^{4,5}.

In addition, government agencies across the world acknowledge the need to address delays in delivering support and resources across social, educational, housing, occupational and health services (e.g., counselling, supportive housing, vocational programmes)^{6–9}. Furthermore, the high incidence of co-occurring mental health conditions among individuals with NDCs underscores the critical need for expanded resources to alleviate the burden on the healthcare system, (i.e. decreased emergency care visits, reducing the incidence of comorbidity)^{10–12}. This has led to growing awareness of a need to develop new models of care and support that can meet the needs of the community. One strategy to facilitate this change is to increase the application of innovative evidence-based technologies, such as AI, to provide personalized support of adaptive functioning skills at scale^{13,14}.

Adaptive functioning, now referred to as functioning, refers to learned skills and behaviours that meet the demands of everyday life¹⁵. People with NDCs often report challenges in socialization¹⁶, communication¹⁷ and daily living activities^{18,19} that impact various parts of life, such as self-care²⁰, employment^{21,22}, education and schooling^{23,24}, mental wellbeing¹², and independent living^{25,26}. Prioritising the promotion of functioning skills

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through novel supports can benefit society more broadly by promoting inclusivity and reducing the current barriers to participating in daily life. According to the WHO²⁷, AI-driven technologies are set to redefine the future of healthcare and promise unparalleled advancements for enhancing patient wellbeing²⁷. Considering this, AI may provide novel opportunities to identify and support functioning needs in individuals with NDCs. AI is often defined as the *science and engineering of making intelligent machines*, consisting of computer systems that can learn, perform, and achieve tasks that typically require human intelligence²⁸. AI encompasses a range of techniques, including autonomous systems²⁹, machine learning³⁰, computer vision³¹, Natural Language Processing (NLP)^{32,33} and more.

The global rise of AI offers an opportunity to develop data-driven approaches to increase access to care, improve personalized therapies and allow for functioning skills to be supported in real time. Medical research has begun to apply AI for mental health^{34,35}, in smart home devices for people who have reduced mobility^{36,37}, and to track physiological biomarkers of behaviour to monitor treatment outcomes^{38,39}. More broadly, AI applications facilitate support in clinical practice. For example, AI tools have been employed alongside medical imaging to support interpretation of scans to aid in diagnosis⁴⁰. They have also been applied to the analysis of patient data to improve tailored assessment and treatments, based on genomic screening, medical history and lifestyle factors⁴¹. Furthermore, AI has been utilized to improve access to healthcare services⁴². In terms of patient functioning, AI has also been implemented to support specific functioning needs, such as the use of smart prosthetics and hearing aids^{43,44}, in speech recognition devices⁴⁵ and in cognitive assistance systems to support daily activities for people living with dementia⁴⁶.

Despite its promise, there remains a lack of benchmarking frameworks to assess the evidence of the use of these technologies²⁷. Moreover, studies can differ on dimensions of efficacy and effectiveness, which often exist on a continuum⁴⁷. These properties have not yet been evaluated systematically in research investigating AI-enabled assistive devices in people with NDCs. Singal et al.⁴⁷ presented criteria upon which studies could be evaluated for efficacy and effectiveness. For the purpose of this review, efficacy is defined as the ability of an intervention to produce a desired outcome under ideal and controlled conditions^{47,48}, whereas effectiveness refers to the intervention's performance in real-world settings where various factors may influence its implementation⁴⁹. These are important to consider, as the opportunities with AI have largely been seen in clinical environments^{50–52}. As a result, the utility of these devices in everyday environments including homes, schools, and workplaces, is notably absent. Additionally, adherence to these interventions, the degree to which individuals follow the agreed-upon duration and incorporate these devices into their daily routine, will be crucial in assessing their long-term effectiveness and integration into daily life⁵³.

To demonstrate that AI assistive devices have utility in the real-world, evidence needs to coincide in everyday settings. AI techniques, such as machine learning, neural networks, autonomous systems, and natural language processing, have great potential to assist as support tools. These data-driven approaches are poised to revolutionize the support of functioning skills in people with NDCs by offering precise support structures that can optimize individual capabilities.

Therefore, this systematic review examines the current state of research on the use of AI-assistive devices to support functioning in this population, taking a particular emphasis on their use in everyday settings. Specifically, this study reviews the current frameworks underpinning the development and implementation of AI-assistive devices for individuals with NDCs, including the range of technologies currently available and how they address various challenges associated with functioning. Through this review, we intend to inform researchers, clinicians, companies and policymakers on the potential benefits and challenges associated with the use of AI-assistive devices in this population.

To date, there has been no review of the literature on AI-assistive devices in a broad range of NDCs across different age groups and from different settings within the community. Furthermore, there has been no

attention on the effectiveness of these interventions specifically for functioning behaviours. The increasing pressures on the healthcare system and unmet needs of individuals with NDCs, in conjunction with the fast proliferation of AI-enabled technology has led to a growing interest in the use of these digital technologies to assist with functioning to improve quality of life. The objective of this review is to evaluate the efficacy and effectiveness of AI-assistive technologies to support functioning in individuals with NDCs. For the purpose of this review, the definition of AI is the science and engineering of making intelligent machines that have the ability to learn, perform and achieve goals²⁸. Within this review and due to the relatively new advancement of AI, we will include studies that report on these AI techniques. Synthesis of quantitative evidence of currently available studies investigating AI-assistive devices will be critically evaluated to assess the outcome of functioning.

Results

A PRISMA flow diagram summarises the systematic search used in this review (see Fig. 1). Titles and abstracts of 4130 papers were reviewed and 15 were found to have met the study's inclusion criteria. The most common reasons for exclusion were having non-assistive device interventions, (i.e. the assistive device that did not use any form AI, 76% of studies).

Study characteristics

Table 1 summarises the characteristics of the included studies in this review. Studies were mainly conducted in the United States (3 studies), the United Kingdom (3 studies) and Spain (3 studies) and published between 2005 and 2022. Overall, studies reported an average of 70% overall male distribution. Eleven studies primarily focused on outcomes in children, two studies in adolescents^{54,55} and two studies focused on outcomes in adults^{56,57}. The most common settings for AI applications were in schools, with one study by Lorenzo et al.⁵⁵ using Interactive Virtual Reality (IVR) to simulate an educational context (i.e. classroom and a playground) and at home. The sample sizes varied across study types, ranging from 4 to 71. To illustrate, Robins et al.⁵⁸ and Nasri et al.⁵⁴ recruited 4 participants in a single-subject study and in a user study, and in contrast, one Randomized Control Trial (RCT) recruited 71 participants⁵⁹.

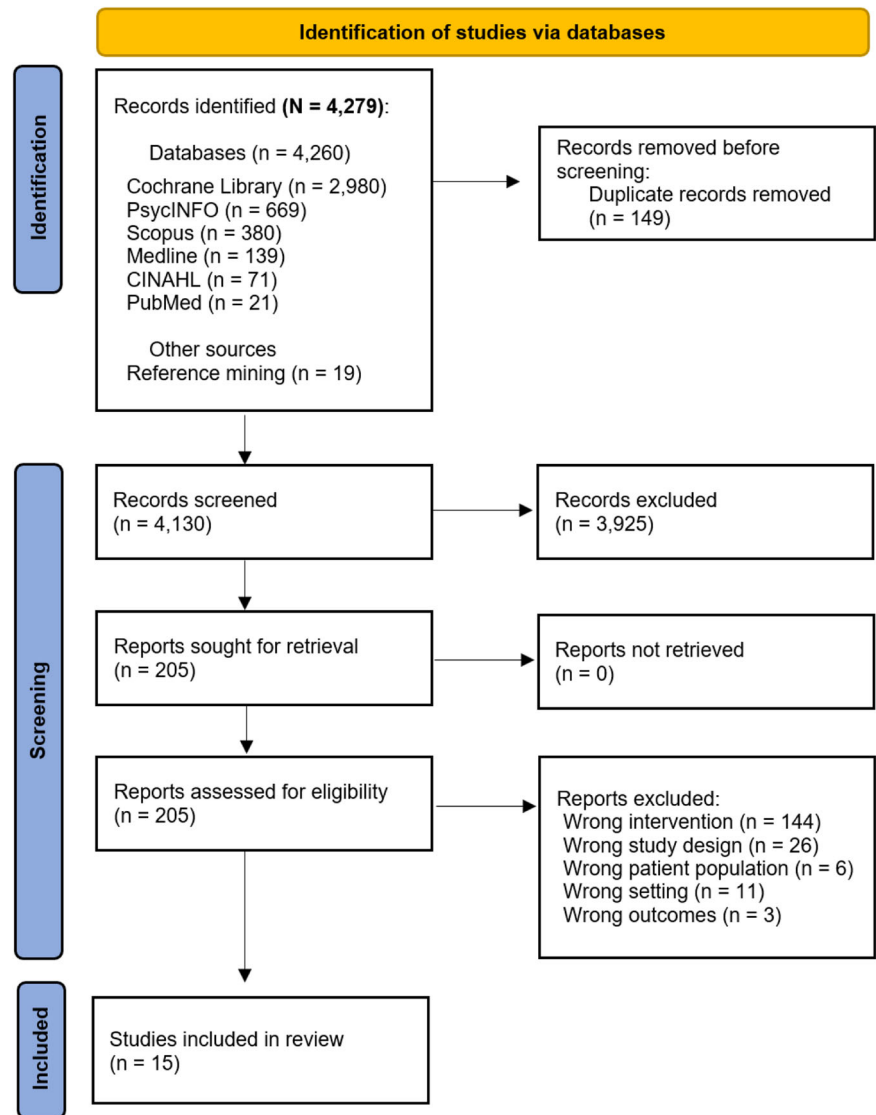
Study design and quality assessment

Overall, three studies in this review were RCTs^{59–61} and the remaining 12 studies adopted quasi-experimental designs^{54–58,62–68}. The JBI Critical Appraisal Checklist for RCTs and Quasi-experimental study designs addressed methodological quality such as randomization procedures, statistical rigour, and completeness of follow up. Based on this tool, the methodological quality of the included studies was found to be adequate with a low risk of bias. A lack of comparison or control groups was the main area where information was insufficient. However, this is not uncommon due to the inherent challenges with finding control groups and appropriate comparators in this population. Five studies were unclear as to whether follow-up was completed. Over half of the studies employed independent researchers or clinicians to carry out assessments.

Participant characteristics, diagnoses and comorbidities

The average age of participants ranged from 4 years to 27 years. Most participants were recruited from the general community and from schools, including inclusive education centers. Recruitment also took place in diagnostic and treatment clinics for people with NDCs. Although NDCs were included broadly in our search strategy, 11 of the 15 (73%) studies included in this review listed ASD as the primary diagnosis of the participants. The other studies listed ADHD, ID, Developmental Coordination Disorder (DCD) and Cerebral Palsy (CP) as the primary diagnosis (see Table 2). Only two studies reported on comorbid diagnoses, namely ID with sensory impairments⁵⁶ and ASD with other psychological and neurological conditions⁵⁹.

Fig. 1 | PRISMA flow diagram. Depiction of screening process and study selection. Note that n indicates the number of studies included at each stage.



Type of technologies and AI

All devices use AI technology or a particular sub-type of AI, such as machine learning. The term 'AI' or 'artificial intelligence' was used as the primary intervention in the included studies ($n = 7$ studies), followed by specific applications such as autonomous systems ($n = 6$ studies), computer vision ($n = 1$ study), machine learning ($n = 1$ study) and predictive modelling ($n = 1$ study) (see Fig. 2). The most common type of technology used was robotics ($n = 6$ studies), followed by applications based on devices such as mobile phones or computers ($n = 3$ studies), a virtual reality system ($n = 2$ studies), tablet devices or screens ($n = 2$ studies) and wearable devices (i.e. Google's 'Superpower Glass') ($n = 2$ studies) (see Fig. 3).

Primary outcome: adaptive functioning

In terms of functioning outcomes, the main functioning outcome that was addressed was social skills ($n = 9$ studies). Communication, motor skills, daily living skills (i.e. self-care activities), emotional skills and other activities such as learning were also investigated. Other single studies did, however, focus on different phenotypes. For example, Medina et al.⁶⁰ recruited an ADHD population and found improved inhibitory control through the use of a mobile equipped with AI, which has been found to be associated with better functioning in this age group (8-9 year olds)⁶⁹.

Summary of findings by functioning outcome

Eleven studies directly reported on the effects of AI-assistive devices for autistic people^{55,57-59,62-68}, the remaining four studies reported on a population of children with ADHD ($n = 1$ study)⁶⁰, DCD ($n = 1$ study)⁶¹, ID ($n = 1$ study)⁵⁶ and cerebral palsy ($n = 1$ study)⁵⁴. These studies enrolled a total of 244 children, 2 adolescents and 19 adults from different community settings. Of these settings, the most frequently cited contexts were in the schools' children attended ($n = 6$ studies), followed by their homes ($n = 4$ studies). A wide variety of interventions were investigated, including robotics, virtual reality systems and wearable and non-wearable devices. Thirteen of the trials had a low risk of bias and two had a medium risk of bias due to unreported randomization procedures or incomplete follow-up^{61,68}. Finally, three studies were identified to have potential funding bias^{59,60,66}. We used the criteria presented by Singal et al.⁴⁷ to determine whether studies could be best characterised as efficacy or effectiveness studies. As described by Singal et al.⁴⁷, most studies used a combination of both approaches. Based on these criteria, 12 of the included studies could largely be classified as efficacy studies (80%)^{54,55,57-64,66,68}, while 3 were best considered effectiveness studies (20%)^{56,65,67}. We note that one study, Bernadini et al.⁶², did not establish efficacy of their AI-assistive intervention (see Tables 3 and 4).

Table 1 | Summary of Included Studies

Author (year)	AI technology	Study design	Sample Size	Post-trial follow-up duration	Experimental setting	Evidence of co-design	Main adaptive functioning outcome/s	Acceptability	Suitability for adoption in daily settings?	Bias
Bernardini et al. ⁶²	ECHOES is a serious game which allows children with ASD to interact with AI virtual characters to practice social and communication skills.	Quasi-experimental (Within-subject)	19	NR	School	No	Socialization and Communication: Social initiations with virtual avatars.	Yes	No	Low
Bimbrawh et al. ⁶³	The COACH device for ASD has specially-built AI software to analyze images in real-time and prompt handwashing actions.	Quasi-experimental (Pilot study)	5	NR	University washroom	No	Daily Living Skills: Improved independent handwashing skills by using an intelligent prompting system.	Yes	Yes	Low
Chung (2021) ⁶⁴	The NAO humanoid robot functions in a closed-loop autonomous mode to provide natural interactions with children.	Quasi-experimental	15	NR	School	No	Socialization: Robotic intervention NAO was successful in enhancing eye contact and verbal initiation.	Yes	Yes	Low
Clabaugh et al. ⁶⁵	Autonomous socially assistive robot 'The Kiwi' provides educational games to reinforce learning activities.	Quasi-experimental (Single subject)	17	NR	Home	No	Daily Living Skills: The Kiwi assistive robot was a successful personalized learning device, particularly in math skills.	Yes	Yes	Low
EbrahimiSani et al. ⁶¹	VR training system of sport games to improve motor skills.	RCT	40	2 months	School	No	Motor skills: Significant improvements in hand rotation, motor planning and control.	Yes	No	Medium
Kohl et al. ⁶⁶	Machine learning-based ABA treatment tracked via a device to create personalized interventions.	Quasi-experimental (Single group)	29	NR	Online	No	Socialization and Communication: Using goal prediction and personalization, ML aids in prioritizing care around specific socio-communication needs.	Yes	Yes	Low
Lancioni et al. ⁵⁶	A smartphone-based technology system equipped with Amazon Alexa to prompt functional activities.	Quasi-experimental (Single subject)	6	NR	Community	No	Daily Living Skills: Simple technology is highly effective in helping people carry out daily tasks independently.	NR	Yes	Low
Lorenzo et al. ⁵⁵	IVR of an educational context with artificially intelligent avatars.	Quasi-experimental	20	NR	VR simulating school settings	Yes	Socialization: Visual strategies, being in a structured environment and completing repetitive tasks improved social skill acquisition.	Yes	Yes	Low
Medina et al. ⁶⁰	KAD_SCL_01 cognitive games with AI.	RCT	29	NR	Home	No	Daily Living Skills: Smart technology showed efficacy in promoting inhibitory control, making everyday activities easier to complete.	NR	NR	Low
Nasri et al. ⁵⁴	Assistive robot LOLA2 equipped with an AI application to reinforce daily living tasks.	Quasi-experimental (User study)	4	NR	Therapy centre	No	Daily Living Skills: Robotic intervention was effective in monitoring and assisting in daily tasks in therapy sessions.	Yes	NR	Low
Washington et al. ⁶⁷	Superpower Glass established by Google Glass uses computer vision.	Quasi-experimental (Pilot study)	14	NR	Home	Yes	Socialization: Repeated use of the device resulted in improvements in social skills, even after intervention was ceased.	Yes	Yes	Low
Robins & Dautenhahn ⁶⁸	Humanoid social robot KASPAR has sensing capabilities that allows autonomous responses.	Quasi-experimental (Case study)	36	NR	School	No	Socialization, Communication and Motor skills: Play scenarios increased social and tactile feedback responses.	Yes	NR	Medium

Table 1 (continued) | Summary of Included Studies

Author (year)	AI technology	Study design	Sample Size	Post-trial follow-up duration	Experimental setting	Evidence of co-design	Main adaptive functioning outcome/s	Acceptability	Suitability for adoption in daily settings?	Bias
Robins et al. ⁵⁸	A humanoid robot 'Robota' that uses autonomous systems to increase basic social interaction skills.	Quasi-experimental (Single subject)	4	NR	School	No	Socialization: Repeated exposure to the robot increased basic social interaction skills, such as imitation and was a shared mediator with others.	Yes	Yes	Low
Silva et al. ⁵⁷	Autonomous Zoomer robotic dog to elicit social communication.	Quasi-experimental (Within-subject)	23	NR	Home or therapy centre	No	Socialization: Appropriate social responses were present, however, not significant when compared to a living dog.	Yes	NR	Low
Voss et al. ⁵⁹	Superpower Glass established by Google Glass uses computer vision.	RCT	71	6 weeks	Home	No	Socialization: Improvements in facial engagement and emotional recognition.	No	Yes	Low

A single-subject study design refers to the effect of an intervention on a single subject's behaviour and response¹⁰⁰. A user study design refers to the effect of an intervention to explore the targeted needs of a specific population¹⁰¹. A within-subject design refers to the same group of participants being exposed to the intervention repeatedly⁹⁷. Post-trial follow up period refers to the follow-up duration that starts after the end of the scheduled trial period⁹⁸. NR not reported, RCT randomized controlled trial, IVR interactive virtual reality, VR virtual reality.

Table 2 | Participant Characteristics

	N	Mean age (years)	Gender (male, %)	Primary diagnosis
Bernardini et al. ⁶²	19	8.5	18 (95)	ASD
Bimbrahw et al. ⁶³	5	5.2	4 (80)	ASD
Chung ⁶⁴	15	8.7	13 (87)	ASD
Clabaugh et al. ⁶⁵	17	NR	15 (88)	ASD
EbrahimiSani et al. ⁶¹	40	8.9	0 (0)	DCD
Kohli et al. ⁶⁶	29	4.1	24 (83)	ASD
Lancioni et al. ⁵⁶	6	NR	1 (16)	ID
Lorenzo et al. ⁵⁵	20	NR	16 (80)	ASD
Medina et al. ⁶⁰	29	9.2	25 (86)	ADHD
Nasri et al. ⁵⁴	4	NR	0 (0)	CP
Washington et al. ⁶⁷	14	9.6	11 (80)	ASD
Robins & Dautenhahn ⁶⁸	36	NR	NR	ASD
Robins et al. ⁵⁸	4	7.8	3 (100)	ASD
Silva et al. ⁵⁷	23	6; 27	21 (91)	ASD
Voss et al. ⁵⁹	71	8.4	63 (89)	ASD

Includes sample size information, mean age, percentage of males and primary diagnosis. NR not reported, ASD autism spectrum disorder, DCD developmental coordination disorder, ID intellectual disabilities, ADHD attention-deficit/hyperactivity disorder, CP cerebral Palsy.

Daily living skills

Of the studies that investigated daily living skills (DLS) (*n* = 5), all found that AI-assistive technologies resulted in improvement in DLS such as handwashing, cleaning, and general household activities. Bimbrahw et al.⁶³ reported improvements in the self-care skill of hand washing in children using AI and computer vision, with prompts presented through a flat-screen monitor. Statistical analysis to confirm the evidence of improvement was, however, not provided⁶³. Parent responses following the intervention were unanimously in favour of using such a device at home, where a device could be used by children to support activities of daily living⁶³. Lancioni et al.⁵⁶ also targeted daily living skills by applying a simple but smart technology in the form of a smartphone, equipped with an Amazon Alexa™ AI system provided to a group of adults with ID with comorbid sensory impairments. The device aimed to assist performing household activities such as cleaning, with prompts occurring 3 to 5 times per day. The mean percentage of activities the participants completed independently increased from below 7 tasks before the intervention to nearly 100 tasks during the intervention⁵⁶. Further information on whether these results can be maintained over time however is needed.

In addition to devices, robotic interventions were also utilized to assist in daily living skills. Nasri et al.⁵⁴ explored whether a robot using an AI-based application could allow for therapists to track how well adolescents with CP could understand and perform tasks such as brushing teeth, mopping and writing on a blackboard. The participants showed enjoyment when interacting with the robot, however, due to cognitive and physical limitations, the AI system could not correctly detect all actions. For example, one participant had to hold themselves up while using the whiteboard due to balance difficulties, which blocked the robot's view⁵⁴. The AI application would therefore require further refinement to detect greater accuracy. In another study investigating daily living skills, Medina et al.⁶⁰ and colleagues examined cognitive stimulation therapy on a mobile, which was controlled by AI to improve inhibitory control performance in 29 children with ADHD. Inhibitory control performance as an outcome was included in this review as it has been established to be associated with better functioning skills⁶⁹. Significant improvements in inhibitory control, as well as cognitive flexibility, working memory and executive functioning, were suggested, and

Fig. 2 | Distribution of AI applications among adaptive functioning outcomes. Representation of AI and its subtypes applied to different functioning outcomes in the included studies. Note that one type of AI may be used to support more than one functioning outcome.

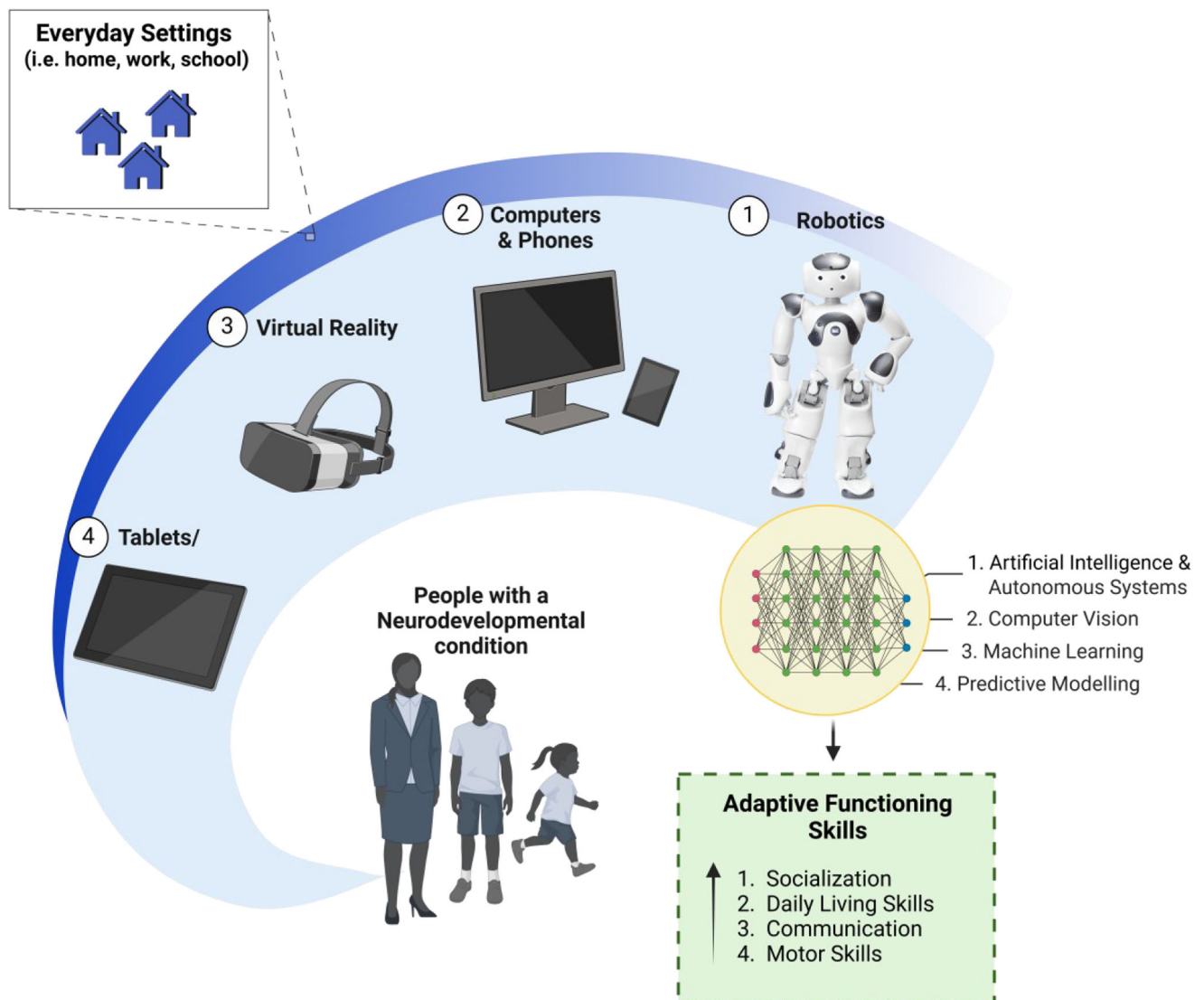
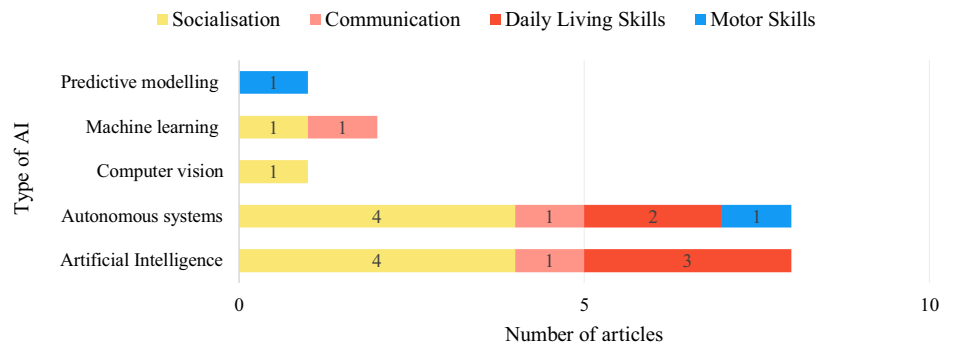


Fig. 3 | Types of technologies, AI interventions and adaptive functioning outcomes. Representation of the most reported types of AI utilized across the most common types of technologies, as well as adaptive functioning outcomes in the included studies. AI-assistive devices, AI, and adaptive functioning outcomes

ordered by frequency (1: Most reported, 4: Least reported) (i.e. 1. Robotics (most reported), 2. Computers & Phones, 3. Virtual reality, 4. Tablets (least reported)). Figure created in BioRender. Saha, S. (2024) <https://BioRender.com/u71z053>.

they made links to similar results found in pharmacological intervention studies. They suggested that digital interventions may provide a promising therapeutic route for executive functions and daily living skills⁶⁰. Additional research with larger samples is necessary to explore this further.

Communication

Studies investigating communication ($n = 3$) found that AI-assistive technologies resulted in improvement in communication skills such as imitation and turn-taking. A study by Bernardini et al.⁶² aimed to facilitate social communication skills through a virtual autonomous agent, but the effect of

Table 3 | Evaluation of study outcomes: effectiveness, efficacy, and adherence

	Effect size measure	Effect size	Effectiveness	Efficacy	Adherence rates (low, medium, high)
Bernardini et al. ⁶²	NR	NR	n/a	No	Medium
Bimbrahw et al. ⁶³	NR	NR	n/a	Yes	High
Chung ⁶⁴	Tau-U (rU)	>0.8	n/a	Yes	High
Clabaugh et al. ⁶⁵	Cohen's <i>d</i>	0.5	Yes	n/a	High
EbrahimiSani et al. ⁶¹	Partial eta squared (η^2)	0.47, 0.65	n/a	Yes	High
Kohli et al. ⁶⁶	NR	NR	n/a	Yes	n/a
Lancioni et al. ⁵⁶	NR	NR	Yes	n/a	High
Lorenzo et al. ⁵⁵	NR	NR	n/a	Yes	Medium
Medina et al. ⁶⁰	Hedges <i>g</i>	0.62	n/a	Yes	Medium
Nasri et al. ⁵⁴	NR	NR	n/a	Yes	Medium
Washington et al. ⁶⁷	NR	NR	Yes	n/a	High
Robins & Dautenhahn ⁶⁸	NR	NR	n/a	Yes	High
Robins et al. ⁵⁸	NR	NR	n/a	Yes	High
Silva et al. ⁵⁷	Partial eta squared (η^2)	0.32, 0.52	n/a	Yes	High
Voss et al. ⁵⁹	NR	NR	n/a	Yes	Low

Includes effect size information, reported effectiveness and/or efficacy, and reported adherence rates. Effect size specifically refers to quantification of observed effect. Effect sizes were quantified using Cohen's *d* and Hedges' *g* to assess the magnitude of group differences, partial eta squared (η^2) to represent variance explained in analysis of variance (ANOVA), and Tau-U to analyze non-overlapping data trends in single-case designs. Efficacy is defined as the ability of an intervention to produce a desired outcome under ideal and controlled conditions^{47,48}. Effectiveness is defined as performance in real-world settings where various factors may influence its implementation. Adherence is defined as the degree to which individuals follow the agreed-upon duration and use of the intervention. NR not reported, n/a not applicable.

Table 4 | Results of effectiveness and efficacy criteria applied to the included studies

	Effectiveness	Efficacy
Bernardini et al. ⁶²		✓
Bimbrahw et al. ⁶³		✓
Chung ⁶⁴		✓
Clabaugh et al. ⁶⁵	✓	
EbrahimiSani et al. ⁶¹		✓
Kohli et al. ⁶⁶		✓
Lancioni et al. ⁵⁶	✓	
Lorenzo et al. ⁵⁵		✓
Medina et al. ⁶⁰		✓
Nasri et al. ⁵⁴		✓
Washington et al. ⁶⁷	✓	
Robins & Dautenhahn ⁶⁸		✓
Robins et al. ⁵⁸		✓
Silva et al. ⁵⁷		✓
Voss et al. ⁵⁹		✓

The effectiveness and efficacy of the included studies were defined according to criteria published by Singal et al.⁴⁷ (see Supplementary Table 5). Studies that scored highest on each of the criteria were grouped accordingly. Most studies (80%) reported efficacy trials, whereas 20% reported effectiveness trials.

the support was not statistically significant in this small sample ($p = >0.05$). The use of a virtual agent to assist in socio-communication strategies for autistic children remains a promising avenue of research⁶². In further examination of communication skills, Kohli et al.⁶⁶ utilized machine learning approaches to Applied Behavioural Analysis (ABA), where personalized treatment goals could be generated and tracked at home through a parent's phone. They found that their machine learning techniques could predict ABA treatment recommendations and goals, with up to 85% accuracy. Personalized goals were argued to contribute to a more effective prioritization of healthcare resources targeted toward functioning needs⁶⁶.

Motor skills

One study examined motor skills as the primary functioning outcome. EbrahimiSani et al.⁷⁰ conducted an investigation on the application of virtual reality training for predictive motor control in children with DCD. Tasks such as hand rotation ($p = 0.015$), action planning ($p = 0.025$) and anticipatory action control ($p < 0.001$) improved significantly with a VR intervention. These motor improvements were also maintained at 2 months follow-up⁷⁰.

Socialization

Social skills were the most common functioning domain targeted in studies investigating AI-assistive devices for people with NDCs. A total of 9 studies targeted this domain (47%). Chung⁶⁴, Robins & Dautenhahn⁶⁸, Robins et al.⁵⁸ and Silva et al.⁵⁷ utilized autonomous systems in social robots and found improvement in children's social behaviours. Multiple studies showed increased eye contact and gaze with a robotic partner, with Chung⁶⁴ finding this increased eye contact was sustained in children after the intervention was withdrawn^{64,68}. Additional improvements in tactile engagement⁶⁸, as well as the emergence of spontaneous and proactive play behaviours⁵⁸ were identified, suggesting the potential benefit of robot-mediated technology in supporting different social skills. Increases in verbal initiation and facial emotional recognition were reported following robotic intervention, social skills that could be considered as common therapy goals for autistic people^{64,71,72}. These findings, however, must be viewed in light of the relatively small sample sizes used. Clabaugh et al.⁶⁵ sought to improve learning skills through personalized instruction and feedback from a Socially Assistive Robot (SAR). They found children to be socially engaged with the robotic device 65% of the time over the period of 1 month⁶⁵. Clabaugh et al.⁶⁵ also reported significant improvement in the targeted skills of learning numerical operations ($p = 0.002$) and mathematical skills ($p < 0.01$) using the SAR device⁶⁵. In contrast, not all findings were supportive of AI-assistive technology at improving social skills, as shown by Silva et al.⁵⁷, who compared living and robotic dogs to test elicitation of social behaviour in children and adults with autism⁵⁷. Silva et al.⁵⁷ found that children showed increased social behaviours in the presence of the living dog, whereas

the robotic dog was found to be just as effective as the living dog in adults⁵⁷.

In one RCT published by Voss et al.⁵⁹, a wearable intervention utilizing AI, known as Superpower Glass, was found to result in significant improvements in social skills as measured by the Vineland Adaptive Behaviour Scale, (VABS-II). In this study, there was an average 4.58 gain in scores on the socialization subscale ($p = 0.005$)⁵⁹. Superpower Glasses were also investigated by Washington et al.⁶⁷ in children with autism. This study used in-built social games to assist in emotional recognition during interactions. They found that repeated use of the Superpower Glasses was associated with an increase in social behaviours, including social interactions and emotional recognition, that remained present even after the device was withdrawn⁶⁷. However, results were mediated by autism symptom severity, such that the greater the number of autism symptoms, the greater the preference for structured social activities (i.e. games to prompt social engagement)⁶⁷. Interestingly, most children in this study preferred both audio and visual feedback, seemingly contradicting the common idea that multiple inputs may cause sensory overload in children with ASD⁷³. One study used AI in a virtual reality environment to assess social skills acquisition⁵⁵. In a virtual reality simulated educational context, 20 children with Asperger's syndrome demonstrated increases in social skill acquisition, as well as elevated motivation and attention, compared to baseline scores⁵⁵. The largest improvements were seen for virtual environments depicting a school playground.

Discussion

This review is the first to systematically evaluate the existing literature on AI-assistive technologies and their effectiveness to support adaptive functioning in individuals with NDCs. We identified 15 studies that utilized AI interventions and showed evidence of utility for children, adolescents and adults with NDCs, including those who had a diagnosis of ASD, ADHD, ID, DCD or CP. Of the functioning outcomes measured, improvements in social skills, daily living activities, communication and motor skills were often reported.

Overall, 14 papers presented in this review supported the use of AI-assistive technologies for improving adaptive functioning outcomes. This was particularly true for social skills and related behaviours (i.e. eye contact frequency, verbal initiation, tactile engagement). AI interventions most commonly used to target social skills were humanoid social robots equipped with autonomous systems. Robins & Dautenhahn et al.⁶⁸ for example, revealed that children with ASD initiated tactile engagement with KASPAR from the first encounter and assisted in teaching children to have an awareness of cause-and-effect behaviours (i.e. 'sad' emotion changes to 'happy' emotion when children interact with the robot)⁶⁸. Social behaviours in children with autism were observed to increase in real-world environments. Clabaugh et al.⁶⁵, for example, found SARs to be an effective personalized intervention for developmental support at home⁶⁵. At-home, personalized learning assistants such as SARs may be an engaging and effective learning assistant, whilst also providing social training across the longer-term⁶⁵. Robins et al.⁵⁸ on the other hand, presented evidence that autonomous robots can effectively enhance social interaction skills in children with autism in school settings⁵⁸. There was only one study that investigated a robotic platform in individuals with CP⁵⁴. This study found it was able to successfully distinguish between daily actions (i.e. brushing teeth, blow-drying hair, mopping the floor) with high accuracy, further reinforcing independent task completion⁵⁴. Chung⁶⁴ revealed that children sustained eye contact frequency and duration with a human instructor after withdrawal of the robotic intervention, highlighting both sustained social skills over time and transferability to human subjects⁶⁴. Finally, in a study looking at elicitation of social behaviours in children and adults with autism using living and robotic dogs, living dogs were found to be more effective at prompting social interactions in children⁵⁷. This suggests that social robotics may have reduced efficacy when compared to their animate counterparts⁵⁷. These studies highlight

the utility of robotics equipped with AI in facilitating functioning, with broader applications to include individualized social skills training and learning task performance strategies.

Other forms of technology, such as wearable devices^{59,67} and Immersive Virtual Environments (IVE) simulating a classroom⁵⁵, revealed improvements in social skills as well as executive functions throughout the duration of the study⁵⁵. This is particularly interesting given evidence that executive function delay is a common concern across most neurodevelopmental conditions^{74,75}. Intriguingly, the study carried out by Voss et al.⁵⁹ found that Superpower Glasses reinforced facial engagement and emotion recognition in children with ASD, which is sometimes regarded as some of the more difficult social behaviours to learn^{59,76,77}. Voss et al.⁵⁹ found low participant adherence to wearing the Superpower Glass intervention, (i.e. participants using the device for half as many days as recommended). Washington et al.⁶⁷, on the other hand, observed certain children could handle multiple sensory inputs (i.e. audio and visual feedback), contradicting the initial concern for sensory overload. The reason for low adherence to the wearable device in Voss et al.⁵⁹, whether due to sensory overload or other factors, needs to be investigated further. The varied adherence rate to wearable technologies suggests that individuals may experience sensory modalities differently and devices will need to be personalized for optimal user experience. Wearable devices such as these offer real-time feedback with personalized guidance that can facilitate broad skill acquisition and independence in daily activities. This may also assist health professionals in delivering support and remote monitoring of behaviours.

While wearable devices can provide real-time feedback, virtual reality immerses users in personalized environments. EbrahimiSani et al.⁶¹ was the only study retrieved that included children diagnosed with DCD in a virtual reality intervention. They found that virtual reality games were effective in improving motor skills in children as it promotes active learning in an immersive space⁶¹. However, the authors emphasized that VR was not a sufficient replacement for actual environments with other children and cautioned against the potential overuse of VR in healthcare practices⁶¹. Along with virtual reality, monitors and smartphones also offered unique interactive features that catered to their users' needs to improve functioning skills. Bimbrahw et al.⁶³ and Lancioni et al.⁵⁶ documented a positive relationship between using their intervention and increases in daily living skills of handwashing and household activities, respectively. These supports had high completion rates by people with autism and ID, showing the benefit of using these tools for more fundamental tasks around the home. However, these findings must be interpreted with caution as small samples were used ($n = 5, 6$)^{56,63}. AI-enabled devices such as these offered tailored experiences that could promote greater independence and hold promise for improving outcomes for people with diverse needs.

There was unanimous support by parents, teachers, and therapists for the use of AI-assistive devices. This feedback was mainly gathered through reports during or after the intervention. For that reason, stronger emphasis on structured co-design practices is needed in the development of novel interventions. The presence of co-design was notably lacking and broader needs of the community need to be addressed concerning the integration of AI-assistive technology in daily activities. Previous research has leveraged co-design practices with AI systems in stroke patients^{78,79} and discussed general assistive technologies with people with ID^{79,80}. However, research has yet to include people with NDCs and their families or support persons in co-design techniques addressing AI-assistive devices to support their everyday functioning.

This review retrieved only two studies^{55,67} that collaborated with the community in their research. Washington et al.⁶⁷ consulted with parents and children, taking an iterative approach in finding patterns of key issues and relevant outcomes that should be addressed⁶⁷. In addition, Lorenzo et al.⁵⁵ carried out monthly interviews with teachers to collect information on behaviour and utility of the IVR avatar intervention over the course of the study. The collection of data from teachers was stated to be integral for students to achieve new skills and adhere to the intervention. The studies in this review would benefit from including more end-users, (e.g. parents,

children, teachers, clinicians), at the beginning of the project and during the design phase, as is utilized in similar research⁸¹. Participatory design is an important area of future research that can ensure the successful design and development of AI technologies for people with NDCs, especially in increasing acceptability and usability of AI tools in the community⁸².

Many technologies addressed here were readily adopted and embraced by individuals with NDCs and their carers. The robotic dog tested by Silva et al.⁵⁷ and exposure to a virtual agent in a VR environment in Bernadini et al.⁶², were the only interventions identified that did not significantly improve functioning skills. This suggests that when compared to that of a living dog, live counterparts remain a potentially more effective tool for eliciting social behaviour⁵⁷. In addition, nuanced facial expressions, body language and audio feedback are important for prompting interactions and facilitating prolonged engagement with technology⁶². Successful adoption of VR was observed in Lorenzo et al.⁵⁵, who recorded increased attention and motivation to continue learning in the IVR environment in primary and secondary school students with Asperger's syndrome⁵⁵. For daily living activities, parents felt that their child was able to perform ADLs better with the device than without it⁶³. Chung⁶⁴ identified increased social motivation in children through a robotic intervention, with a learning curve revealing a slow initial acquisition of the device followed by a steep acceleration of learning⁶⁴.

Feedback from teachers and therapists on social robots conveyed the perception that these interventions could help promote initiative-taking and further develop a sense of self in autistic children⁶⁸. Furthermore, parents reported that Superpower Glasses assisted in improving social interactions and emotion recognition at home, a noticeable change in behaviour which continued when not wearing the device⁶⁷. In contrast, the acceptability of the Superpower Glasses tested in Voss et al.⁵⁹ was less embraced, with participants using the device for half as many days as recommended⁵⁹. Overall, most of the population showed engagement with the AI-technology interventions and parents and teachers supported the increased interest in developing these tools for therapeutic interventions.

All papers in this review reported the use of assistive technologies in an everyday environment. This was to evaluate interventions that might have real-world relevance and ecological validity. Investigating assistive technologies within the community can assist in better tailoring interventions to a person's unique needs and capabilities. In this sense, AI is the ideal tool to use as it can quickly learn and adapt to its user's needs. The technologies discussed in this review have great potential to be used in the community, however, the implementation of AI-assistive devices investigated in these studies requires further research. The results suggest that technologies that are customizable, used in a familiar setting and utilize transdisciplinary approaches in their implementation (i.e. researchers and teachers) have the greatest potential at being streamlined into everyday settings^{55,63,83}. Many of the papers conclude that their findings can be applied to clinical and educational settings^{55,64}. Nevertheless, a significant lack of evidence-based targets when testing AI-assistive devices is apparent, which is needed to ensure that there is empirical evidence that functioning outcomes are improving. Only two studies^{56,59} in this review used a standardized assessment of functioning, the VABS-2. Other studies relied mainly on video coding behaviours. Although video coding is a valid measurement technique, greater emphasis on evidence-based functioning outcomes is essential to ensure that researchers can accurately evaluate the performance of AI-assistive devices against established benchmarks that can be compared across studies and populations. Importantly, including clear benchmarks in the form of standardized assessments will translate more effectively to clinical practice and aid in informing more effective therapeutic interventions and policy frameworks.

This review takes a methodological approach to extracting and critically appraising relevant papers to summarize the effectiveness of AI interventions. It has included research that has recruited participants from community settings and incorporated user studies that have not been extensively explored in this area before using systematic review methods. Finally, the reliability of these findings is strengthened by the involvement of

two reviewers to improve the integrity of these results and reduce bias in reporting. The primary limitation of this study is the limited number of studies identified, many of which involve relatively small sample sizes. This reduces the overall generalizability of the current study's findings, as results gathered from small samples may not adequately represent the broader population. As a result, these insights may not fully capture the complexities of AI-assistive device interventions in neurodiverse populations and reduce the ability to draw broader conclusions. In addition, this study did not consider business model factors and related to this, the funding required to disseminate different applications. Such factors have potential to further limit generalizability, particularly in lower-resource communities. There is also a scarcity of research investigating these interventions in females and adults which is particularly concerning given the substantial social, economic and personal support needs for adults and females with NDCs. In addition, there was no evidence of any AI application that met our search criteria for people with communication disorders (e.g. Speech Sound disorder), Tic disorders (e.g. Tourette's disorder) or other NDCs (e.g. Global Developmental Delay). Adult users and people with different comorbidities were less represented, highlighting the limited evidence within these groups and the tendency for research to focus on children with a single primary diagnosis. These limitations highlight the numerous untapped avenues for further pursuit and it is hoped that this review can stimulate and promote more studies to fill these many gaps. The recent development of these tools requires various methods of testing and improvement before they can be adopted in the community. First, longitudinal research is required to be carried out to test the effectiveness of the devices in everyday contexts⁶⁸. Secondly, other participant groups need to be involved especially in consideration of the nature of comorbidity in NDCs, as well as the scarcity of evidence in female populations and adults⁸⁴. Transdiagnostic targets that cut across all neurodevelopmental conditions may have the greatest potential impact for social benefit. Finally, although multiple quantitative studies investigated these devices, including RCTs, the results of these select few can't be used to draw significant conclusions on their effect in the community. Therefore, further investigation using RCT designs is warranted, as well as the adoption of qualitative approaches to ascertain community feedback, such as co-design techniques.

In the pursuit of accessible and effective support options, the future of this area of research will undoubtedly turn to cutting-edge technologies to target functioning skills. Based on our findings, there is a need to develop frameworks for the implementation of AI-assistive technologies as the field progresses. To support their uptake, a focus on implementing these technologies across local settings is required by engaging at the micro, meso, and macro levels^{85–87}.

To achieve this, the first step involves capturing nuanced micro-level experiences of individuals with NDCs within their communities. These insights are often gathered through qualitative, co-design approaches to guide engagement with individuals and their support network, to understand practical constraints, design requirements, and perceived utility of AI technology in everyday settings throughout its development. For example, Washington et al.⁶⁷ utilized qualitative interviews with parents to evaluate the Superpower Glass intervention to assist future designs of similar systems⁶⁷. Capturing the daily experiences of people with NDCs is useful to do through the lens of The WHO's International Classification on Disability, Health and Functioning (ICF) framework, where health and disability is viewed alongside relevant personal and environmental factors⁸⁸. The WHO's classification approach offers a more holistic view to identify the use of AI-assistive devices. To advance this, insights should also be gathered during the design and development phases^{89–91}. This is essential for enhancing the utility of AI technologies early in their life cycle, ensuring that they can be optimized for real-world deployment from the outset.

Secondly, meso-level approaches must be adaptable to bridge micro and macro-level activities. Meso-level approaches are required to be highly flexible, capturing both valuable community-level information whilst

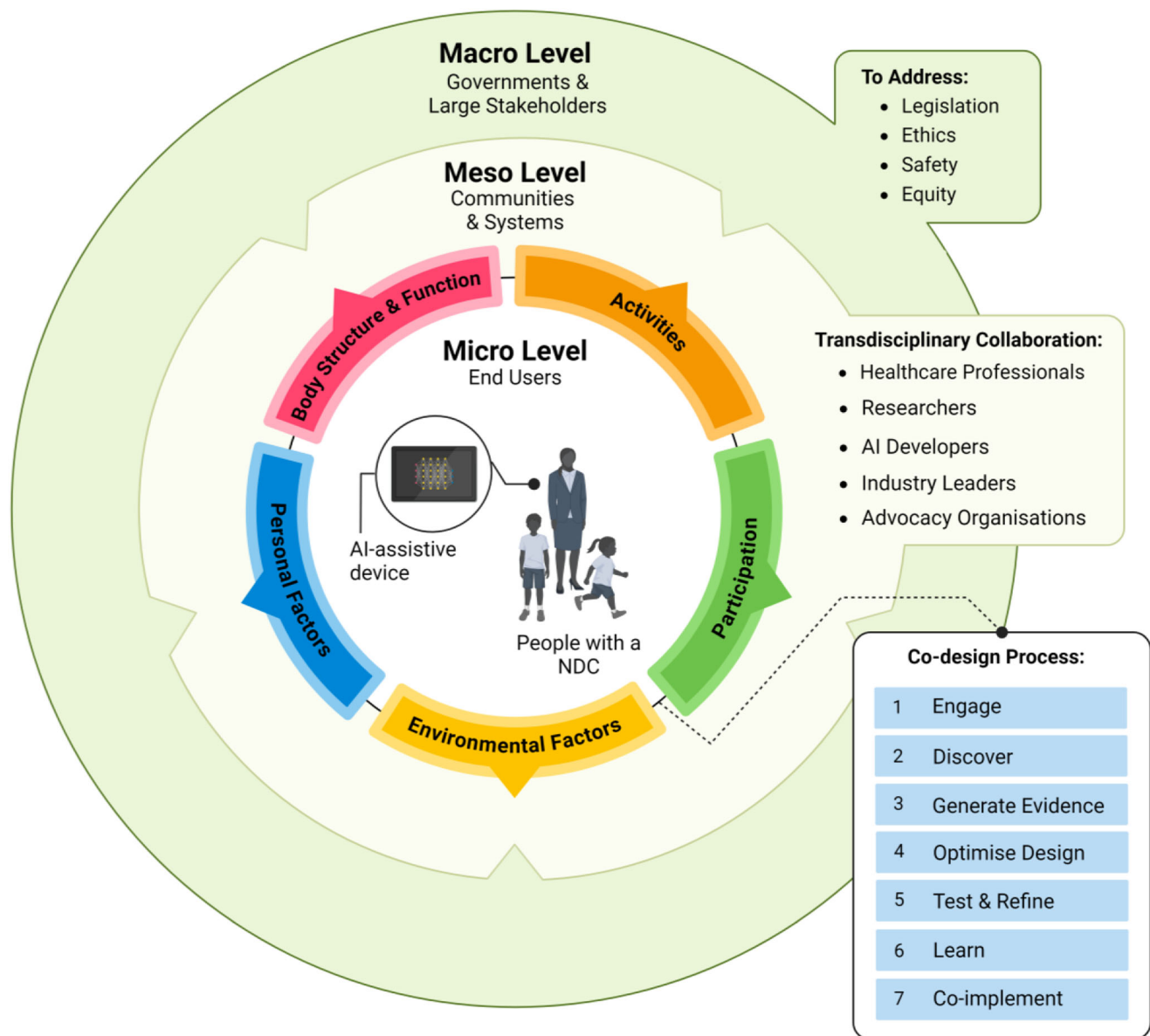


Fig. 4 | Framework for implementing AI-driven interventions for people with neurodevelopmental conditions. Micro: End-user collaboration is adapted from the WHO's International Classification on Disability, Health and Functioning (ICF) framework⁸⁸. These experiences are gathered through co-design to optimize AI technology and user experience whilst fostering community empowerment^{103,104}.

Meso: Transdisciplinary partnerships amongst specialists enable the capture of community and systems-level information. Macro: larger government agencies and stakeholders have the authority to address large-scale challenges to implementation and facilitate broader adoption of AI-assistive technology. Figure created in BioRender. Saha, S. (2024) <https://BioRender.com/y76a702>.

driving systems-level change to enable broad societal impact⁹². For the implementation of AI-assistive tools for people with NDCs, meso-level guidance comes from active collaboration among healthcare professionals, such as neurodevelopment assessment specialists, researchers, industry leaders, advocacy organisations, and AI developers. Implementing this step into practice involves AI developers and researchers partnering with end-users to optimise AI device design and accessibility. Clinicians and researchers should coordinate to recommend ways AI can perform effective healthcare strategies. One example of a meso-level approach can be identified in the study by EbrahimiSani et al.⁶¹, who collaborated closely with trained therapists after incorporating feedback from parents. This collaboration facilitated the successful delivery of personalized AI interventions to participants⁶¹.

Finally, macro-level approaches address wider implementation issues at the systemic level. This involves significant ethical, regulatory and resource considerations that need to be recognised by governments and

large stakeholders^{92,93}. Addressing broad challenges surrounding implementing novel technologies requires addressing ethical concerns (e.g. privacy and consent), ensuring equitable access within the community (e.g., funding models), and providing legislative structure that guarantees that these tools are delivered in a safe and compassionate manner^{94,95}. None of the included studies in this review reported direct impact on policy initiatives, however, future research should consider conducting joint research ventures to closely align research goals with policy needs and engage in public discourse to enhance broader translation of findings.

The exchange of information between micro-level communities and macro-level entities is crucial for ensuring equitable and sustainable change. This cyclical relationship allows for grassroots insights to inform policy initiatives, which can in turn better provide for local needs (see Fig. 4). By maintaining this flow of information, the development of AI-assistive technologies can be adjusted to foster positive outcomes across a diverse population with various functioning needs.

In addition, it can ensure that AI technologies continue to have a positive impact on society, remaining inclusive, responsive and capable of driving systemic improvements. This framework establishes a foundation for integrating novel technologies into society and serves as a benchmark for applying AI-assistive tools in other areas of healthcare, such as mental health. The future of AI technology is in addressing transdiagnostic needs to reduce ongoing healthcare burdens, thereby easing pressure on the entire healthcare system⁹⁶.

In conclusion, this is the first review to systematically evaluate the current literature on the effectiveness of AI-assistive technologies to support adaptive functioning in people with NDCs, taking a particular focus on the use of these devices in daily settings. The results of this review suggest that AI-driven supports can improve functioning in areas of socialization, daily living, communication and motor skills. Overall, robotics equipped with AI were the most used technologies to improve functioning skills in people with NDCs, followed by smartphones or computers, virtual reality, tablets and wearable devices. These technologies have been viewed positively by people with NDCs, as well as their families and teachers. There does remain a lack of evidence to support the adoption of these devices into everyday settings, highlighting the need for ongoing research in this field. To ensure the ongoing success of these interventions in the future, larger cohorts of people with different NDCs and targeting transdiagnostic supports should be explored. Additionally, research directed at adult and female populations is needed. Finally, emphasis on active collaboration between researchers and the community in developing and evaluating AI-assistive devices is necessary. By prioritising these factors, AI-assistive devices have a greater chance of successfully integrating into people's everyday lives to improve their overall functioning and wellbeing.

Methods and materials

This review was registered with PROSPERO (CRD42024500675). The search was undertaken in accordance with the guidelines outlined by the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA)⁹⁷.

Study selection and eligibility criteria

A three-step search strategy was utilized by first conducting a limited search of MEDLINE, PubMed and Scopus to identify articles on the topic. The keywords contained in the titles and abstracts of relevant articles were used to develop a full search strategy (see Supplementary Table 1). The search strategy was adapted for each included database. Only studies published in English were included. Studies published since 2000 were included due to the unlikely development of AI technology before this date.

The searches were conducted in April 2024 and references collated and uploaded to Endnote citation software where duplicates were removed⁹⁸. The search revealed $N = 4278$ papers and the results of the search are presented in a PRISMA flow diagram (see Fig. 1). One company was contacted regarding the specific type of AI used in their device. Authors of the included papers were contacted for any unpublished data. This review considered papers that reported on the delivery of an AI-assistive device through any method (e.g. virtual reality, tablets, robotics), as well as the use of AI-assistive devices as co-interventions to behavioural therapy. This review considered studies where the main outcome was adaptive functioning, such as activities of daily living, self-care (e.g. dressing, bathing), mobility (e.g. independent activities), socialization and communication. In addition, it also considered papers that reported data on relevant psychological outcomes. This review focused on studies that reported on functioning outcomes, taking particular focus in daily contexts such as in educational, vocational, work or community settings. Experimental studies, such as RCTs were included. Due to the relatively recent introduction of these technologies and the nature of the population, we considered case studies, pilot studies and exploratory studies that utilized quantitative measures to gain insight into the different characteristics in which functioning can be supported (see Supplementary Table 2).

Data extraction

The title and abstracts of selected citations were assessed in detail against the inclusion criteria by three independent reviewers (NP/CS/RK). Reasons for exclusion of papers at full-text screening were reported. Data was extracted from studies included in the full-text review by two independent reviewers (NP and CS/MM) using the standardized JBI data extraction tools⁹⁹. Data was extracted independently to reduce selection bias and minimize error. The data extracted included specific details about the participants, study design, interventions, and outcomes of interest to the review, including diagnoses, type of AI and device used and functioning outcomes. Any disagreements between reviewers at each stage of the selection process were resolved through discussion until mutual consensus was reached.

Quality assessment

Eligible studies were critically appraised by two independent reviewers (NP and CS/MM) using the standardized critical appraisal instrument from JBI for experimental or quasi-experimental studies (see Supplementary Table 3). Studies were rated 'high' to 'low' in terms of the risk of bias. We examined factors relating to treatment groups, reliability of the methods used, if follow-up was completed and the statistical appropriateness of the methods used, among others. The results of the critical appraisal are reported in narrative form and in Supplementary Table 4 (see Supplementary). All studies, regardless of their methodological quality, underwent data extraction and synthesis. Following critical appraisal, studies that did not meet the quality threshold were excluded based on the joint agreement among reviewers.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Code availability

Not applicable.

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References

1. WHO, W. H. O. *Disability*, <<https://www.who.int/news-room/fact-sheets/detail/disability-and-health>> (2023).
2. Maenner, M. J. et al. Prevalence and characteristics of autism spectrum disorder among children aged 8 years — autism and developmental disabilities monitoring network, 11 sites, United States, 2020. *MMWR Surveill. Summ.* 2023 **72**, 1–14 (2020).
3. UN, U. N. In *Treaty Series* 2515 Vol. 3 (ed United Nations) 37 (2006).
4. Boulton, K. A. et al. Access and barriers to supports for children and caregivers attending public child developmental assessment services: findings from the Sydney child neurodevelopment research registry. *Autism Res.* **17**, 555–567 (2024).
5. Boulton, K. A. et al. Diagnostic delay in children with neurodevelopmental conditions attending a publicly funded developmental assessment service: findings from the Sydney Child Neurodevelopment Research Registry. *BMJ Open* **13**, e069500 (2023).
6. Krahn, G. L., Walker, D. K. & Correa-De-Araujo, R. Persons with disabilities as an unrecognized health disparity population. *Am. J. Public Health* **105**, S198–S206 (2015).
7. HHS, U. S. D. O. H. A. H. S. Vol. 2024-09237 (ed U.S. Department of Health and Human Services) 40066-40195 (U.S. Department of Health and Human Services, Washington, D.C., 2024).
8. Government, A. (ed Department of Social Services) 72 (Department of Social Services, Canberra, ACT, 2021).
9. Canada, G. O. (ed Accessible Canada Act (the Act)) 92 (Government of Canada, Ottawa, Canada, 2019).

10. Vohra, R., Madhavan, S. & Sambamoorthi, U. Emergency department use among adults with autism spectrum disorders (ASD). *J. Autism Dev. Disord.* **46**, 1441–1454 (2016).
11. Lineberry, S. et al. Co-occurring mental illness and behavioral support needs in adults with intellectual and developmental disabilities. *Community Ment. Health J.* **59**, 1119–1128 (2023).
12. Boulton, K. A. et al. Mental health concerns in children with neurodevelopmental conditions attending a developmental assessment service. *J. Affect Disord.* **335**, 264–272 (2023).
13. Malbon, E., Carey, G. & Meltzer, A. Personalisation schemes in social care: are they growing social and health inequalities? *BMC Public Health* **19**, 805 (2019).
14. Cardonha, C., Britto Mattos, A. & Laiola Guimarães, R. *A Platform to Support Personalized Training of People with Disabilities*. (2016).
15. Tillmann, J. et al. Investigating the factors underlying adaptive functioning in autism in the EU-AIMS Longitudinal European Autism Project. *Autism Res.* **12**, 645–657 (2019).
16. DePape, A.-M. & Lindsay, S. Lived experiences from the perspective of individuals with autism spectrum disorder: a qualitative meta-synthesis. *Focus Autism Other Dev. Disabil.* **31**, 60–71 (2016).
17. Tager-Flusberg, H. & Kasari, C. Minimally verbal school-aged children with autism spectrum disorder: the neglected end of the spectrum. *Autism Res.* **6**, 468–478 (2013).
18. Blanco-Martínez, N. et al. Participation in everyday activities of children with and without neurodevelopmental disorders: a cross-sectional study in Spain. *Children*. **7**. <https://doi.org/10.3390/children7100157> (2020).
19. Perry, N. et al. A psychometric investigation of health-related quality of life measures for paediatric neurodevelopment assessment: reliability and concurrent validity of the PEDS-QL, CHU-9D, and the EQ-5D-Y. *Autism Res.* <https://doi.org/10.1002/aur.3127> (2024).
20. Hakobyan, A. & Harutyunyan, Z. The importance of self-care skills development for children with autistic spectrum disorder from the parents' perspective. *Armen. J. Spec. Educ.* **4**, 21–38 (2021).
21. Winsor, J. et al. StateData: the national report on employment services and outcomes. (University of Massachusetts Boston, Institute for Community Inclusion, Boston, MA, 2018).
22. FitzGerald, E., DiRezze, B., Banfield, L., Nichol, G. K. & Rosenbaum, P. A scoping review of the contextual factors impacting employment in neurodevelopmental disorders. *Curr. Dev. Disord. Rep.* **8**, 142–151 (2021).
23. Nadeau, M. F., Massé, L., Argumedes, M. & Verret, C. Education for students with neurodevelopmental disabilities-Resources and educational adjustments. *Handb. Clin. Neurol.* **174**, 369–378 (2020).
24. DuPaul, G. J. & Langberg, J. M. In *Attention-deficit hyperactivity disorder: A handbook for diagnosis and treatment*, 4th ed. 169–190 (The Guilford Press, 2015).
25. Antolini, G. & Colizzi, M. Where do neurodevelopmental disorders go? casting the eye away from childhood towards adulthood. *Healthcare*. **11**. <https://doi.org/10.3390/healthcare11071015> (2023).
26. Sullivan, A. & Smith, T. Facilitators and barriers that impact the transition to adulthood for young adults with autism spectrum disorder (ASD). *Arch. Phys. Med. Rehabil.* **102**, e105 (2021).
27. WHO, W. H. O. Generating evidence for artificial intelligence-based medical devices: a framework for training, validation and evaluation. (World Health Organization, Geneva, 2021).
28. McCarthy, J. What is Artificial Intelligence? *Artificial Intelligence Definitions*. <https://hai.stanford.edu/sites/default/files/2020-09/AI-Definitions-HAI.pdf> (2007).
29. Bryson, J. & Winfield, A. Standardizing ethical design for artificial intelligence and autonomous systems. *Computer* **50**, 116–119 (2017).
30. Helm, J. M. et al. Machine learning and artificial intelligence: definitions, applications, and future directions. *Curr. Rev. Musculoskelet. Med.* **13**, 69–76 (2020).
31. Ettalibi, A., Elouadi, A. & Mansour, A. AI and computer vision-based real-time quality control: a review of industrial applications. *Procedia Comput. Sci.* **231**, 212–220 (2024).
32. Khurana, D., Koli, A., Khatter, K. & Singh, S. Natural language processing: state of the art, current trends and challenges. *Multimed. Tools Appl.* **82**, 3713–3744 (2023).
33. Sarker, I. H. AI-based modeling: techniques, applications and research issues towards automation, intelligent and smart systems. *SN Comput. Sci.* **3**, 158 (2022).
34. Khawaja, Z. & Bélisle-Pipon, J. C. Your robot therapist is not your therapist: understanding the role of AI-powered mental health chatbots. *Front. Digit. Health* **5**, 1278186 (2023).
35. D'Alfonso, S. et al. Artificial intelligence-assisted online social therapy for youth mental health. *Front. Psychol.* **8**. <https://doi.org/10.3389/fpsyg.2017.00796> (2017).
36. Facchinetti, G., Petrucci, G., Albanesi, B., De Marinis, M. G. & Piredda, M. Can smart home technologies help older adults manage their chronic condition? a systematic literature review. *Int. J. Environ. Res. Public Health* **20**. <https://doi.org/10.3390/ijerph20021205> (2023).
37. Hussein, A., Adda, M., Atieh, M. & Fahs, W. Smart home design for disabled people based on neural networks. *Procedia Comput. Sci.* **37**, 117–126 (2014).
38. Romanowicz, M. et al. Machine learning identifies smartwatch-based physiological biomarker for predicting disruptive behavior in children: a feasibility study. *J. Child Adolesc. Psychopharmacol.* **33**, 387–392 (2023).
39. Saliba, M. et al. PISTACHIo (Preemption of diSruptive behAvior in CHildren): real-time monitoring of sleep and behavior of children 3–7 years old receiving parent-child interaction therapy augment with artificial intelligence - the study protocol, pilot study. *Pilot Feasibility Stud.* **9**, 23 (2023).
40. Ardila, D. et al. End-to-end lung cancer screening with three-dimensional deep learning on low-dose chest computed tomography. *Nat. Med.* **25**, 954–961 (2019).
41. Topol, E. J. High-performance medicine: the convergence of human and artificial intelligence. *Nat. Med.* **25**, 44–56 (2019).
42. Sim, I. et al. Clinical decision support systems for the practice of evidence-based medicine. *J. Am. Med. Inf. Assoc.* **8**, 527–534 (2001).
43. Simon, A. M., Hargrove, L. J., Lock, B. A. & Kuiken, T. A. Target Achievement Control Test: evaluating real-time myoelectric pattern-recognition control of multifunctional upper-limb prostheses. *J. Rehabil. Res. Dev.* **48**, 619–627 (2011).
44. Lesica, N. A. et al. Harnessing the power of artificial intelligence to transform hearing healthcare and research. *Nat. Mach. Intell.* **3**, 840–849 (2021).
45. Zdravkova, K., Krasniqi, V., Dalipi, F. & Ferati, M. Cutting-edge communication and learning assistive technologies for disabled children: an artificial intelligence perspective. *Front. Artif. Intel.* **5**. <https://doi.org/10.3389/frai.2022.970430> (2022).
46. Ienca, M., Wangmo, T., Jotterand, F., Kressig, R. W. & Elger, B. Ethical design of intelligent assistive technologies for dementia: a descriptive review. *Sci. Eng. Ethics* **24**, 1035–1055 (2018).
47. Singal, A. G., Higgins, P. D. & Waljee, A. K. A primer on effectiveness and efficacy trials. *Clin. Transl. Gastroenterol.* **5**, e45 (2014).
48. Gartlehner, G., H. R., Nissman, D. *Criteria for distinguishing effectiveness from efficacy trials in systematic reviews*. Agency for Healthcare Research and Quality (US), Vol. 12 (2006).
49. Godwin, M. et al. Pragmatic controlled clinical trials in primary care: the struggle between external and internal validity. *BMC Med. Res. Methodol.* **3**, 28 (2003).

50. Karalis, V. D. The integration of artificial intelligence into clinical practice. *Appl. Biosci.* **3**, 14–44 (2024).
51. Krishnan, G. et al. Artificial intelligence in clinical medicine: catalyzing a sustainable global healthcare paradigm. *Front Artif. Intell.* **6**, 1227091 (2023).
52. Takata, K. et al. Social skills training using multiple humanoid robots for individuals with autism spectrum conditions. *Front Psychiatry* **14**, 1168837 (2023).
53. Chakrabarti, S. What's in a name? Compliance, adherence and concordance in chronic psychiatric disorders. *World J. Psychiatry* **4**, 30–36 (2014).
54. Nasri, N. et al. Assistive robot with an AI-based application for the reinforcement of activities of daily living: technical validation with users affected by neurodevelopmental disorders. *Appl. Sci.* **12**, 9566 (2022).
55. Lorenzo, G., Pomares, J. & Lledo, A. Inclusion of immersive virtual learning environments and visual control systems to support the learning of students with Asperger syndrome. *Comput. Educ.* **62**, 88–101 (2013).
56. Lancioni, G. E. et al. People with intellectual and sensory disabilities can independently start and perform functional daily activities with the support of simple technology. *PLoS ONE*. **17**, ArtID e0269793 (2022) <https://doi.org/10.1371/journal.pone.0269793> (2022).
57. Silva, K., Lima, M., Santos-Magalhaes, A., Fafiaes, C. & de Sousa, L. Living and robotic dogs as elicitors of social communication behavior and regulated emotional responding in individuals with autism and severe language delay: a preliminary comparative study. *Anthrozoos* **32**, 23–33 (2019).
58. Robins, B., Dautenhahn, K., Boekhorst, R. T. & Billard, A. Robotic assistants in therapy and education of children with autism: can a small humanoid robot help encourage social interaction skills? *Univers. Access Inf. Soc.* **4**, 105–120 (2005).
59. Voss, C. et al. Effect of wearable digital intervention for improving socialization in children with autism spectrum disorder: a randomized clinical trial. *Jama, Pediatr.* **173**, 446–454 (2019).
60. Medina, R. et al. Electrophysiological brain changes associated with cognitive improvement in a pediatric attention deficit hyperactivity disorder digital artificial intelligence-driven intervention: randomized controlled trial. *J. Med. Internet Res.* **23**, e25466 (2021).
61. EbrahimiSani, Sohrabi, M., Taheri, H., Agdasi, M. T. & Amiri, S. Effects of virtual reality training intervention on predictive motor control of children with DCD-A randomized controlled trial. *Res. Dev. Disabil.* **107**, 107 (2020).
62. Bernardini, S., Porayska-Pomsta, K. & Smith, T. J. ECHOES: an intelligent serious game for fostering social communication in children with autism. *Inf. Sci.* **264**, 41–60 (2014).
63. Bimbrahw, J., Boger, J. & Mihailidis, A. Investigating the efficacy of a computerized prompting device to assist children with autism spectrum disorder with activities of daily living. *Assistive Technol.* **24**, 286–298 (2012).
64. Chung, E. Y. h. Robot-mediated social skill intervention programme for children with autism spectrum disorder: an ABA time-series study. *Int. J. Soc. Robot.* **13**, 1095–1107 (2021).
65. Clabaugh, C. et al. Long-term personalization of an in-home socially assistive robot for children with autism spectrum disorders. *Front. Robot AI* **6**, 110 (2019).
66. Kohli, M., Kar, A. K., Bangalore, A. & Ap, P. Machine learning-based ABA treatment recommendation and personalization for autism spectrum disorder: an exploratory study. *Brain Inform.* **9**. <https://doi.org/10.1186/s40708-022-00164-6> (2022).
67. Washington, P. et al. SuperpowerGlass: a wearable aid for the at-home therapy of children with autism. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* Vol. 1, Article 112. <https://doi.org/10.1145/3130977> (2017).
68. Robins, B. & Dautenhahn, K. Tactile interactions with a humanoid robot: novel play scenario implementations with children with autism. *Int. J. Soc. Robot.* **6**, 397–415 (2014).
69. Vuontela, V. et al. Working memory, attention, inhibition, and their relation to adaptive functioning and behavioral/emotional symptoms in school-aged children. *Child Psychiatry Hum. Dev.* **44**, 105–122 (2013).
70. EbrahimiSani, S., Sohrabi, M., Taheri, H., Agdasi, M. T. & Amiri, S. Effects of virtual reality training intervention on predictive motor control of children with DCD - a randomized controlled trial. *Res. Dev. Disabil.* **107**, 103768 (2020).
71. Carmo, J. C., Duarte, E., Pinho, S., Marques, J. F. & Filipe, C. N. Verbal fluency as a function of time in autism spectrum disorder: an impairment of initiation processes? *J. Clin. Exp. Neuropsychol.* **37**, 710–721 (2015).
72. Loth, E. et al. Facial expression recognition as a candidate marker for autism spectrum disorder: how frequent and severe are deficits? *Mol. Autism* **9**, 7 (2018).
73. Asperger, H. *'Autistic psychopathy' in childhood*. (Cambridge University Press, 1991).
74. Sadozai, A. K. et al. Executive function in children with neurodevelopmental conditions: a systematic review and meta-analysis. *Nat. Human Behav.* <https://doi.org/10.1038/s41562-024-02000-9> (2024).
75. Lee, D. et al. Attention and executive delays in early childhood: a meta-analysis of neurodevelopmental conditions. <https://doi.org/10.1038/s41380-024-02802-3> (2024)
76. Landowska, A. et al. Automatic emotion recognition in children with autism: a systematic literature review. *Sensors*. **22**. <https://doi.org/10.3390/s22041649> (2022).
77. Corbett, B. A. et al. A functional and structural study of emotion and face processing in children with autism. *Psychiatry Res* **173**, 196–205 (2009).
78. Ventura, S. et al. Co-designing an interactive artificial intelligent system with post-stroke patients and caregivers to augment the lost abilities and improve their quality of life: a human-centric approach. *Front. Public Health* **11**, 1227748 (2023).
79. Woodward, K., Kanjo, E., Brown, D. J., McGinnity, T. M. & Harold, G. In the hands of users with intellectual disabilities: co-designing tangible user interfaces for mental wellbeing. *Personal. Ubiquitous Comput.* **27**, 2171–2191 (2023).
80. Sitbon, L. & Farhin, S. Co-designing interactive applications with adults with intellectual disability: a case study. in *Proc. of the 29th Australian Conference on Computer-Human Interaction* (2017).
81. Jibb, L. A. et al. Parent and clinician perceptions and recommendations on a pediatric cancer pain management app: a qualitative co-design study. *PLOS Digit Health* **2**, e0000169 (2023).
82. Torous, J., Nicholas, J., Larsen, M. E., Firth, J. & Christensen, H. Clinical review of user engagement with mental health smartphone apps: evidence, theory and improvements. *Evid. Based Ment. Health* **21**, 116–119 (2018).
83. Cerga-Pashoja, A., Gaete, J., Shishkova, A. & Jordanova, V. Improving reading in adolescents and adults with high-functioning autism through an assistive technology tool: a cross-over multinational study. *Front. Psychiatry*. **10**, ArtID 546 (2019). <https://doi.org/10.3389/fpsy.2019.00546>.
84. Taheri, A., Meghdari, A., Alemi, M. & Pouretamad, H. Human-robot interaction in autism treatment: a case study on three pairs of autistic children as twins, siblings, and classmates. *Int. J. Soc. Robot.* **10**, 93–113 (2018).
85. Gama, F. et al. Implementation frameworks for artificial intelligence translation into health care practice: scoping review. *J. Med. Internet Res.* **24**, e32215 (2022).
86. Sawatzky, R. et al. Implications of response shift for micro-, meso-, and macro-level healthcare decision-making using results of

- patient-reported outcome measures. *Qual. Life Res.* **30**, 3343–3357 (2021).
87. Capano, G. & Toth, F. Health policy under the microscope: a micro policy design perspective. *Front. Public Health* **11**. <https://doi.org/10.3389/fpubh.2023.1180836>. (2023)
 88. WHO, W. H. O. (ed World Health Organisation) (World Health Organisation, New York City, United States, 2001).
 89. Sreenivasan, A. & Suresh, M. Design thinking and artificial intelligence: a systematic literature review exploring synergies. *Int. J. Innov. Stud.* **8**, 297–312 (2024).
 90. Nicosia, F. M. et al. Using multiple qualitative methods to inform intervention development: Improving functional status measurement for older veterans in primary care settings. *PLoS One* **18**, e0290741 (2023).
 91. Gonzales, A., Custodio, R., Lapitan, M. C. & Ladia, M. A. End users' perspectives on the quality and design of mHealth technologies during the COVID-19 pandemic in the philippines: qualitative study. *JMIR Form. Res.* **7**, e41838 (2023).
 92. Hampton, M., O'Hara, S. & Gearin, E. Assessing restorative community development frameworks—a meso-level and micro-level integrated approach. *Sustainability* **16**, 2061 (2024).
 93. Masefield, S. C., Msosa, A., Chinguwo, F. K. & Grugel, J. Stakeholder engagement in the health policy process in a low income country: a qualitative study of stakeholder perceptions of the challenges to effective inclusion in Malawi. *BMC Health Serv. Res.* **21**, 984 (2021).
 94. Elliott, K. et al. Towards an equitable digital society: artificial intelligence (AI) and CORPORATE DIGITAL RESPONSIBILITY (CDR). *Society* **58**, 179–188 (2021).
 95. Shneiderman, B. Bridging the gap between ethics and practice: guidelines for reliable, safe, and trustworthy human-centered AI systems. *ACM Trans. Interact. Intell. Syst.* **10**, 26. <https://doi.org/10.1145/3419764>. (2020)
 96. Hickie, I. B. Moving beyond stepped care to staged care using a novel, technology-enabled care model for youth mental health. *Med. J. Aust.* **211**, 404–405 (2019).
 97. Moher, D., Liberati, A., Tetzlaff, J. & Altman, D. G. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med.* **6**, e1000097 (2009).
 98. EndNote (Clarivate, Philadelphia, PA, 2013).
 99. Barker, T. H. et al. The revised JBI critical appraisal tool for the assessment of risk of bias for randomized controlled trials. *JBI Evid. Synth.* **21**, 494–506 (2023).
 100. Hersen, M. & Barlow, D. H. *Single case experimental designs: Strategies for studying behavior change*. (Pergamon, 1976).
 101. Lyon, A. R. & Koerner, K. User-centered design for psychosocial intervention development and implementation. *Clin. Psychol.* **23**, 180–200 (2016).
 102. Llewellyn-Bennett, R., Bowman, L. & Bulbulia, R. Post-trial follow-up methodology in large randomized controlled trials: a systematic review protocol. *Syst. Rev.* **5**, 214 (2016).
 103. Government, N. (ed Agency for Clinical Innovation) (NSW Government, Canberra, ACT, 2024).
 104. Slattery, P., Saeri, A. K. & Bragge, P. Research co-design in health: a rapid overview of reviews. *Health Res. Policy Syst.* **18**, 17 (2020).

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Author contributions

Authors N.P., C.S., K.B., and A.J.G. designed the research study, authors N.P., M.M., and C.S. conducted title and abstract screening, full-text screening, data extraction and critical appraisal. Authors K.B. and A.J.G. contributed to analysis and interpretation. All authors reviewed drafts and contributed to the final version of the submitted manuscript.

Competing interests

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Additional information

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